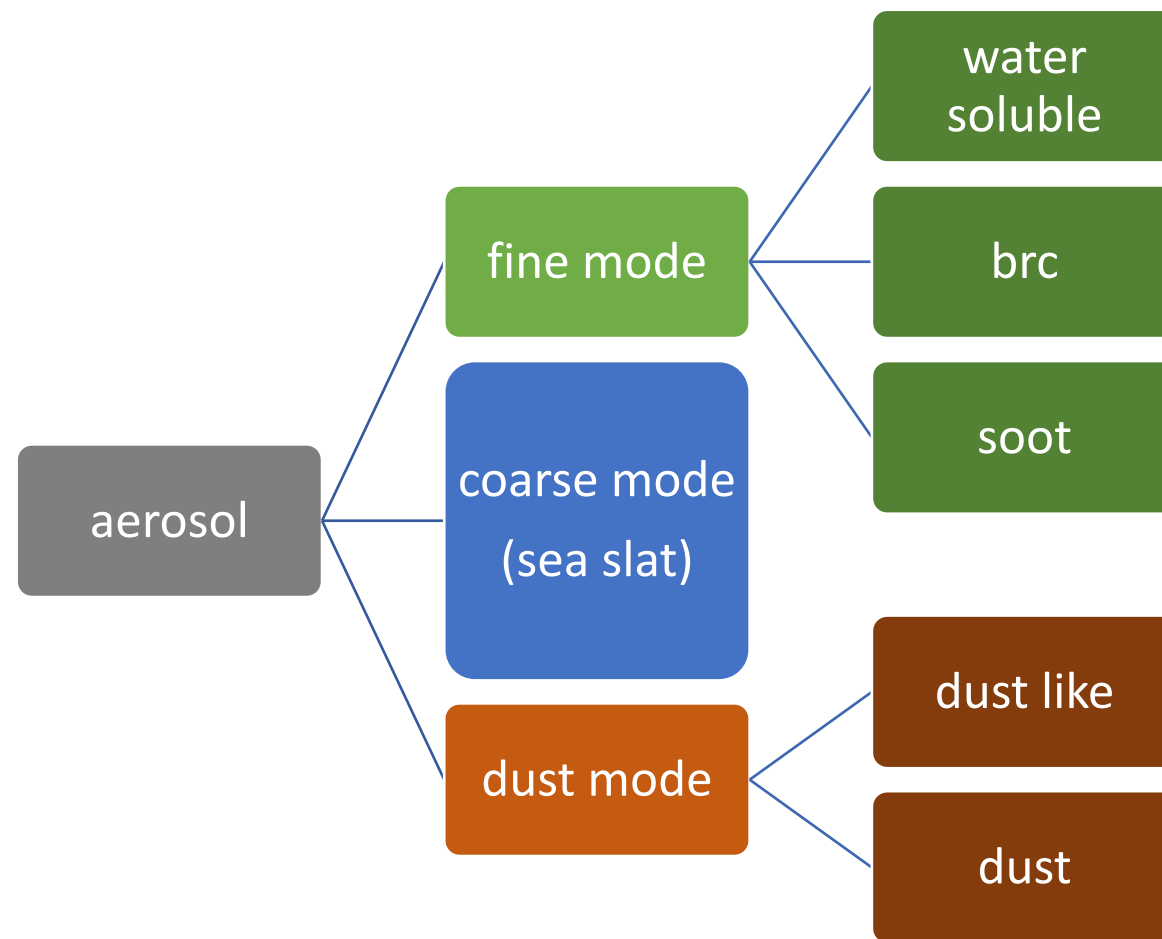
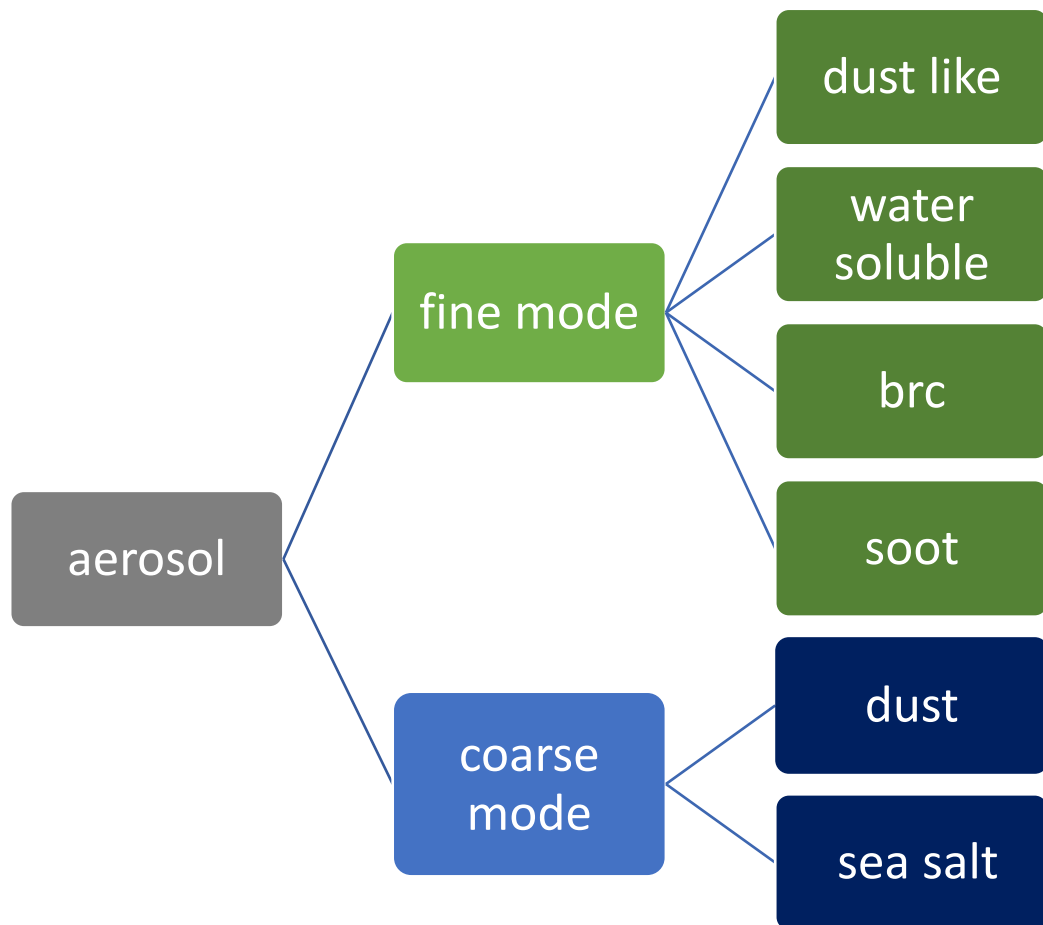
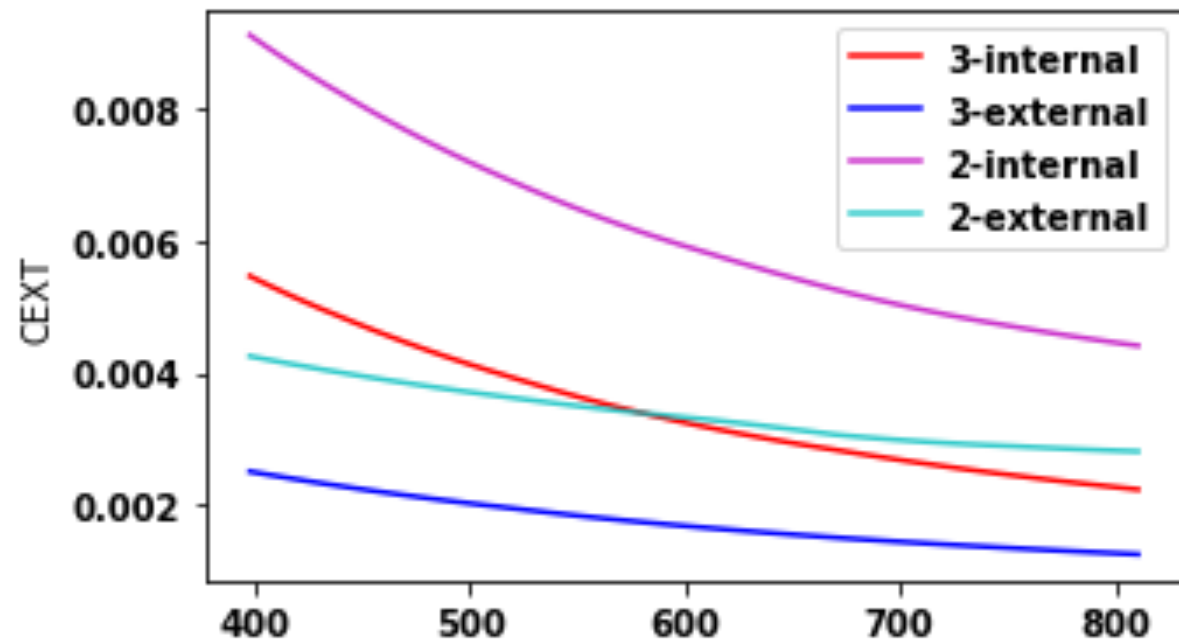


Aerosol Model V.7

(Update)

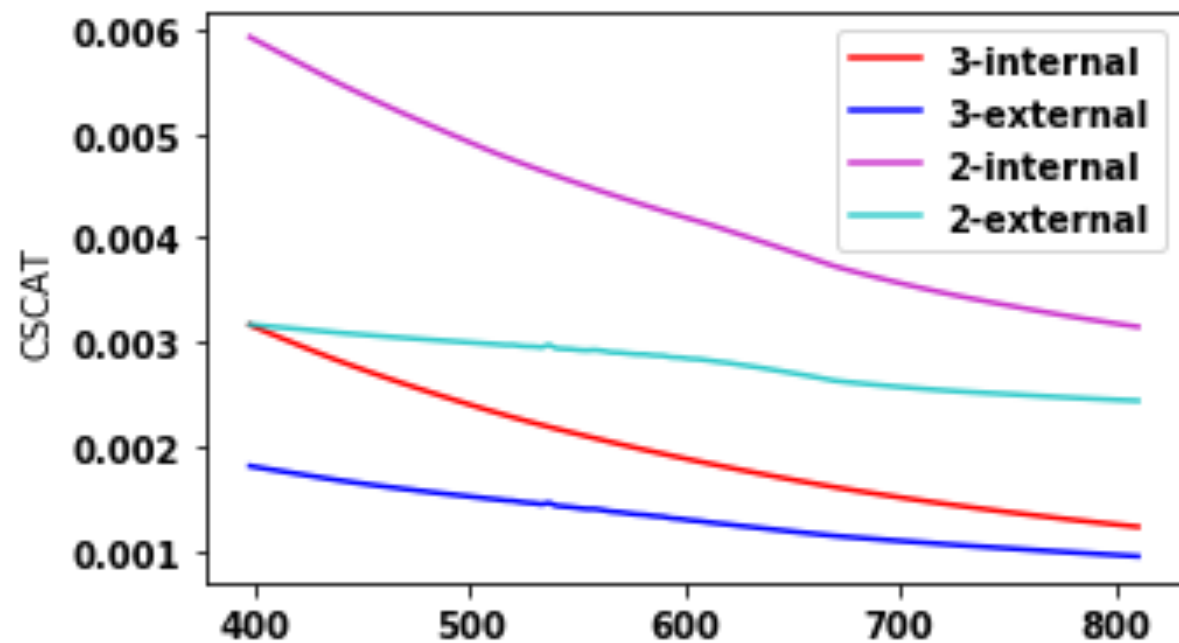
Fine mode internal mixing and external mixing
Fine mode, coarse mode and dust mode aerosols



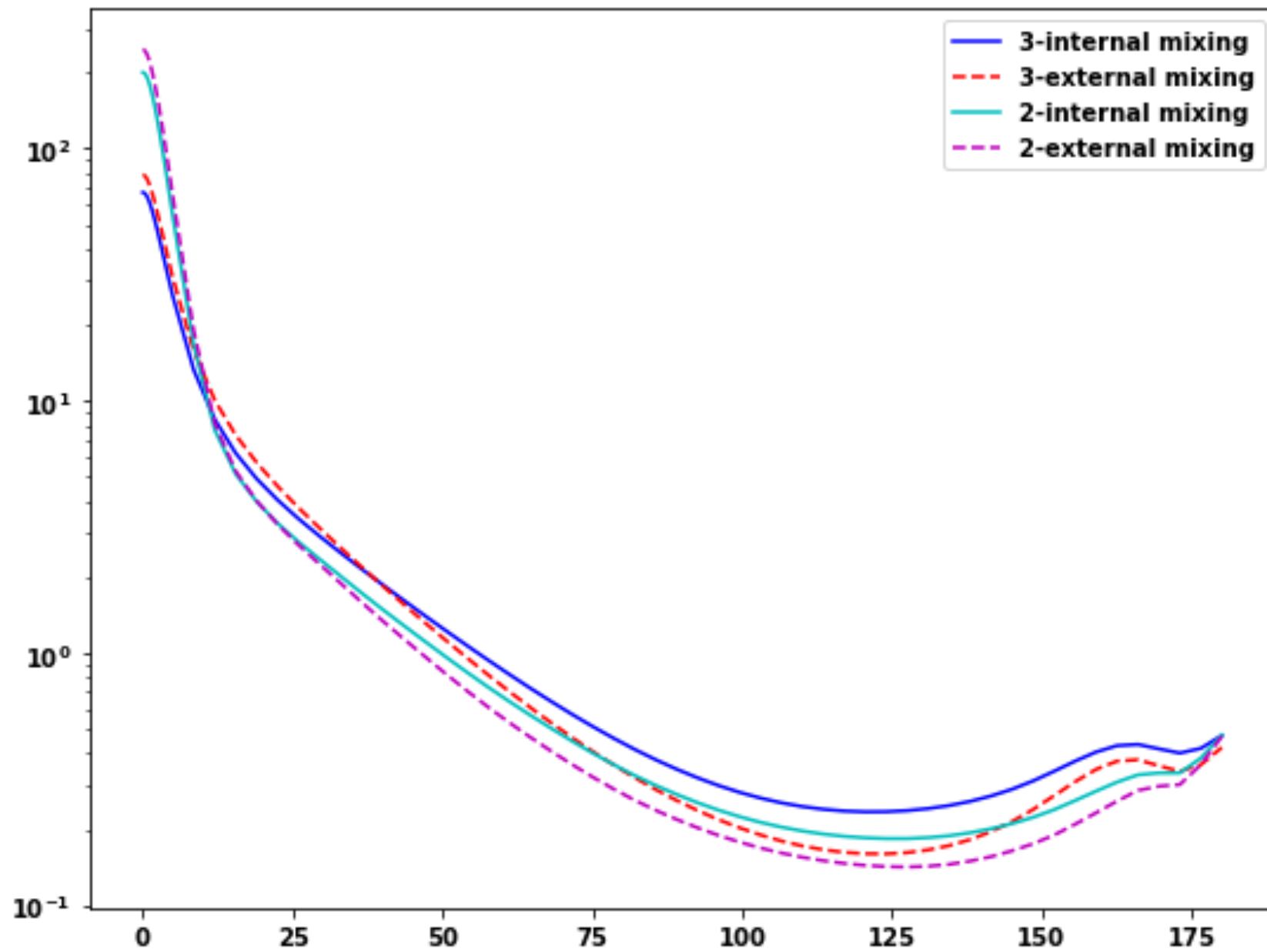


3 – number of aerosol modes = 3

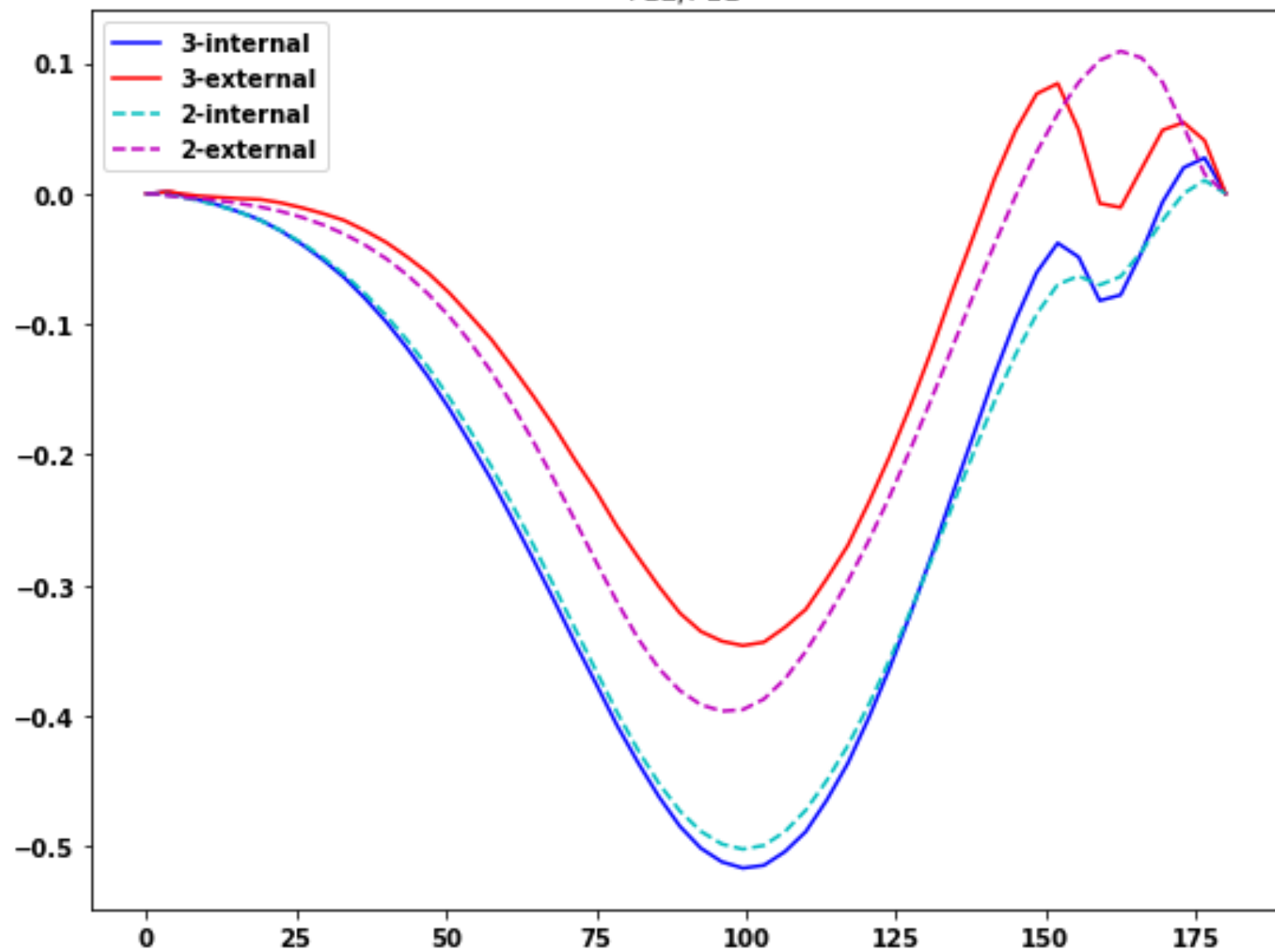
2 – number of aerosol modes = 2



P11



P12/P11



nmode=3

```
21      #IAEROSOL=1,20. 1-10 is Shettle and Fenn, 11-20 Ahmad model, 21 flexible
1      #flexibility flag - defines whether the flexible aerosol model is selected
3      #number of modes flag bi-modal=2, tri-modal=3
0.500000      # fine mode fraction: by volume
0.200000 #dust fraction
0.600000 #spherical fraction in dust mode
0.600000      #rf2 water soluble contribution in fine mode: by volume
0.100000      #rf3 brown carbon contribution in fine mode:by volume
2      #Mixing of fine mode aerosols external = 2, internal = 1, internal with zia = 0
2      # dust variance select 1-5 from 1.2, 1.5, 2. , 2.5, 3. (data base)
15     # dust effective radius select 1-15 from 0.2, 0.4, 0.6, 0.8, 1. , 1.2, 1.4, 1.6, 1.8,2.,2.2, 2.4, 2.6,2.8, 3.
0.600000      # relative humidity < 1.0
0.150000      # dry radius fine mode spherical - internal mixing by volume
2.019400      # dry radius coarse mode spherical (sea salt) by volume
0.806000      # standard deviation fine mode
0.672000      # standard deviation coarse mode
0.200000      # dry radius dust like (fine mode) by volume
0.150000      # dry radius water soluble (fine mode) by volume
0.300000      # dry radius brown carbon (fine mode) volume
0.050000      # dry radius soot (fine mode) volume
2.019400      # dry radius sea salt (coarse mode) volume
0      #wv_seg_flag, 0: all; 1: seg1+3only; 2: seg2only; 3 seg1+2+3; 4: seg4only
output_flexible__nmode3_fmixing_2_fmfrac_0.500_dustfrac_0.200_dsfrac_0.600_fwatersol_0.600_fBrc_0.100_fsoot_0.300_t2
```

nmode=2

```
21      #IAEROSOL=1,20. 1-10 is Shettle and Fenn, 11-20 Ahmad model, 21 flexible
1      #flexibility flag - defines whether the flexible aerosol model is selected
2      #number of modes flag bi-modal=2, tri-modal=3
0.500000      # fine mode fraction: by volume
0.200000      #spherical fraction in coarse mode: by volume
0.100000      #rf1 dust like contribution in fine mode: by volume
0.600000      #rf2 water soluble contribution in fine mode: by volume
0.100000      #rf3 brown carbon contribution in fine mode:by volume
2      #Mixing of fine mode aerosols external = 2, internal = 1, internal with zia = 0
2      # dust variance select 1-5 from 1.2, 1.5, 2. , 2.5, 3. (data base)
15      # dust effective radius select 1-15 from 0.2, 0.4, 0.6, 0.8, 1. , 1.2, 1.4, 1.6, 1.8,2.,2.2, 2.4, 2.6,2.8, 3.
0.600000      # relative humidity < 1.0
0.150000      # dry radius fine mode spherical - internal mixing by volume
2.019400      # dry radius coarse mode spherical (sea salt) by volume
0.806000      # standard deviation fine mode
0.672000      # standard deviation coarse mode
0.200000      # dry radius dust like (fine mode) by volume
0.150000      # dry radius water soluble (fine mode) by volume
0.300000      # dry radius brown carbon (fine mode) volume
0.050000      # dry radius soot (fine mode) volume
2.019400      # dry radius sea salt (coarse mode) volume
0      #wv_seg_flag, 0: all; 1: seg1+3only; 2: seg2only; 3 seg1+2+3; 4: seg4only
output_flexible__nmode2_fmixing_2_fmfrac_0.500_cmsfrac_0.200_fdust_0.100_fwatersol_0.600_fBrc_0.100_fsoot_0.200_t2
```